

A Home Energy Story

Every home has a story. When my family and I moved into our home in 1989, and for the next several years thereafter, we received all of our electricity from the power grid and paid hefty utility bills. Now we give back to the grid and pay only a small amount to the utilities. Our split-level, 2,700 ft² home in San Jose, California houses our family of two grade school children, a stay-at-home mom, and me, a businessman dad. The four-bedroom, three-bath home met the required energy standards at the time, which included insulation in all the perimeter walls, floors, and ceilings and double-glazed windows. We had two 72% efficient natural gas furnaces and a gas water heater. Cooking and laundry used electricity exclusively. In addition, we installed insulation within the ceilings and within the walls between rooms, primarily to make the home quieter.

Over the years we did several things to reduce our energy consumption. We put solar-reflective film on our south- and west-facing windows. We changed out nearly all our incandescent bulbs for fluorescent lights or light-emitting diode (LED) lights. We added an attic fan above the bedrooms and upstairs ceiling fans to reduce the air conditioning load. In May 2003 we replaced the furnaces with a 96.6% efficient condensing unit downstairs and an 82% efficient unit upstairs (a more efficient model wouldn't fit in the heater closet). We also added electronic filters to the furnaces to clean the recirculated air.

In August 2003 we installed a 9.5 kW PV system on the south (back) side of our roof. The relatively large system consists of 72 BP Solar 3160 panels and 4 Sunny-boy 2500 inverters, each with a Sunny Breeze heat sink fan. The PV panels are mounted on the roof using Tile Trac. The mounting system has posts for connecting



CLINTON KENNEDY

When Clinton Kennedy and his family moved into their home in 1989, and for the next several years thereafter, they received all of their electricity from the power grid and paid hefty utility bills.

to or through roofing materials, rails that mount to the posts, and clips that are used to bolt the solar panels into place.

Listen Up, Contractors

The process of selecting and specifying a system was complicated. In an ideal world, the contractors would always be educating the buyers, since the buyers will go through the process a few times, at most, but contractors go through it many times. My contractor was a good installer but, in my opinion, a horrible communicator. In spite of my best efforts, he has never fully delivered on his contract. He still owes me as-built drawings and one or two other small things. One of his references warned me of this, but I thought I knew better.

For me, success for the contractor and the client boils down to a few simple factors:

- Select at least three contractors to bid and listen carefully to what their ref-

erences say. Don't think a shortcoming stated in a reference is something you can overcome. Discard that contractor and select another.

- Any roof surface that receives six or more hours a day of direct unshaded sunlight is suitable for a PV system. In general, the better the angle of the sun, the better the power generation.

- A significant part of the cost of a solar system is in its infrastructure—breakers, switches, inverters, conduit, bidirectional electric meter, permits and inspections, mounting rails, and the labor required to install these components. You get the best payback by installing as many solar panels as your inverter(s) can handle.

- Most contractors will specify a system that will give you the best economic payback. You may want to maximize coverage on your useful roof surfaces in order to contribute electricity to the power grid; to allow yourself capacity to convert gas-burning appliances to electricity; or to

add new appliances, like a hot tub, or a swimming pool. You might even hope to recharge an electric vehicle some day.

At the time our system was installed, the state of California had a rebate and a

ifornia income tax credit. Combined, the cost was reduced by about 59%, so we chose to maximize coverage on our useful roof surfaces. By doing this, we felt we would have sufficient capacity to contribute to the grid and to change or add electric alternatives to natural-gas-burning appliances. We also felt that over the 25–30 year useful life of the system, current and rising energy prices would more than offset the capital cost. In fact, we thought our cost would be recovered in around 12 years.

A Different Angle on Payback

A break-even analysis like the one above distracts most potential purchasers from the real economic benefits of installing a solar system. In a recent article in the *Los Angeles Times* (March 13, 2005), Barbara E. Hernandez describes a 2 kW system that cost \$15,000 and yielded average monthly utility savings of \$150. It may seem like a

lot of energy is being used if the homeowners can save that much each month. But after we moved into our house, and before we installed our PV system in

2003, we consumed an average of 1,000–1,230 kWh per month, at a cost that ranged from \$0.12/kWh in 1990 to \$0.19/kWh in 2002. When financed over 30 years at 6% interest as part of a home mortgage, the monthly loan payments for the home described in the *Los Angeles Times* article go up \$90 but are offset by the \$150 in utility savings, allowing the purchasers to put \$60 per month in their pockets.

Our PV system has now been in operation for 19 months. At the end of the first 12 months, our cumulative electricity production was 3,713 kWh more than our consumption. That power went to the grid without economic benefit to us, but the payback is the satisfaction of knowing that we offset a little capacity coming from the highly polluting peaker plants (see “Meter Matters”). Since our system was turned on, due to net-metering and time-of-use electricity pricing, we have paid nothing for electricity off the grid; we only pay \$0.19 per day as a connectivity fee—basically a charge for reading the meter each month.

A Surplus from the Sun

We would like to use some of our excess electricity to offset or eliminate our natural gas consumption, further lowering our energy bills and cutting back even more on the greenhouse gas emissions that our family is responsible for—more payback that doesn’t appear in the accountant’s bottom line. We started two projects. The first project involved purchasing 1,500W electric space heaters for each of our four bedrooms. The second will involve purchasing an energy efficient electric water heater.

The benefit we see in using electric space heaters is that they put the heat where and when we need it, whereas using a furnace means heating both used and unused rooms. We purchased four Honeywell HZ-519 electric baseboard heaters (about \$50 each) and used them extensively in two of the bedrooms. The heaters provided more than adequate heat in these rooms and limited the use



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The author decided to purchase an electric water heater and electric space heater in order to use up some of the excess electricity generated by his PV system.

tax credit available. The California Energy Commission rebate was \$4 per watt of generating capacity. The net amount was then eligible for a 15% Cal-

METER MATTERS

I am using a bidirectional time-of-use meter on the Pacific Gas and Electric Company (PG&E) E-7 rate structure. With the time-of-use meter you are actually generating power and getting a credit of \$0.325/kWh from noon to 6 pm during the sunniest six months of the year, and buying power back at about \$0.085 during other hours, weekends, and holidays. I had about \$1,000 credit at the end of the first year (July 22, 2004). PG&E zeroed the credit and kept the money, which is part of the

net-metering agreement. I received no economic benefit from the excess power, and that was at least part of my motivation for putting more of my energy use on electricity.

To have a little fun, I wrote a letter to the president of PG&E pointing out the amount of my true-up credit. I acknowledged our net metering deal, and I asked when I could expect his thank-you letter for the \$1,000 I contributed to his company. He had his assistant write a nice letter to me, so I concluded that he was a good sport.

of our gas furnaces. (We had them in the other two bedrooms as well, where visitors occasionally used them.) One of the heaters failed and one seems to produce less heat than it did when it was new, so we suspect that these portable units last only a year. If we decide to make baseboard heaters a permanent solution, we will install appropriate, long-lasting units.

After weeding out other offsetting factors, we think we used about the same amount of electricity this year as we did at this time last year, though our combined gas and electricity consumption was less. Our last child moved out in April 2004, so we undoubtedly saved electricity after that. A murkier factor is energy saved by reducing the use of our high-efficiency gas furnaces. I say murky because the high-efficiency furnaces blow air longer, and have another fan in their exhaust vents, and our furnaces have electronic air filters.

Our next project is installing an electric tank-type water heater. We first looked into tankless gas and electric water heaters. With tankless heaters, you save the standby energy costs related to keeping a tank of water hot, but you pay a premium when the unit is actually heating water. To rapidly heat up water flowing at a rate of several gallons per minute takes a whole lot of electricity or gas. But, overall, you can save energy with tankless units (see "Tankless Option Improving," p. 18).

Given our excess supply of electricity, I focused on electric tankless units, but I found that I would probably have to upgrade my electric service capacity in order to have enough wattage to heat water. Tank-type electric water heaters need roughly a 240V, 30-amp electric circuit. A tankless unit can require double that amount. Either circuit will



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more than likely bypass electric subpanels and come directly from the main power panel outside the home. Older homes may need to upgrade the capacity of their main power panels even for a tank-type electric heater. This can mean removing and replacing the main power panel and wires back to the public utility—a complicated endeavor.

I moved to researching electric tank-type water heaters and found that most information comes from parts of the country rich in hydroelectric power. Most of the energy cooperatives recommended the Marathon line of water heaters from Rheem, because they purportedly use less electricity, are better

insulated, and have no metal tank to corrode.

There are several factors to keep in mind when replacing a gas water heater with electric:

- You need more storage capacity. Electric heaters take several times as long as gas heaters to heat a tank of water. For our needs, the factor was about 60% more. To be more than safe, we chose a 105-gallon heater to replace our 50-gallon gas heater.

- You need to reinforce the heater stand and add seismic bracing. More water means a fatter and taller tank and more weight. At about 8.3 lb per gallon, we will have to support a heater weighing more than 900 lb.

- You need to have a properly installed electric circuit. This is likely to be a 240V, 30-amp line with a breaker and a heater side cutoff switch.

- You might need a time clock. If you are on time-of-use electric metering, you may want the heater to be off during peak periods.

- You might need to cap the vent pipe, since it will no longer be used.

My hope is that by this time next year I can report that my solar power system is supplying all our own home energy needs over a 12-month period. If I am successful, we will no longer be

burning natural gas and we will be supplying as much as we take from the electricity grid. The story of our house continues, but it's close to happily ever after—at least when we look at saving energy and helping to clean up the environment.



—Clinton Kennedy

Clinton Kennedy is a plumber's kid who became a Navy electronics technician and then earned college degrees in business and accounting. For the last 27 years, he has managed service businesses in the Santa Clara valley and remains intrigued with the ways that technology can be used to improve our environment and maintain our quality of life.